

Secured Wireless Network for Aircraft Intra Communication

Peter Reji

Dept of Aerospace Engineering
International Institute of Aerospace
Engineering and Management
(IIAEM), Jain University
Bangalore, India
rrpeter@gmail.com

Dr K Natarajan

Dept of Aerospace Engineering
International Institute of Aerospace
Engineering and Management
(IIAEM), Jain University
Bangalore, India
k.natarajan@jainuniversity.ac.in

Dr. Shobha K R

Dept of Electronics and
Telecommunication Engineering
Ramiah Institute of Technology (RIT)
Bangalore, India
shobha_shankar@msrit.edu

Abstract—Aviation industry has been looking at wireless technology as an option to establish data network for avionics to reduce the weight and in turn reduce operational and maintenance costs. However, introducing the wireless network in an aircraft brings to the front, the issue of security. Intra communication using wireless can make aircraft systems vulnerable to manipulations by an adversary. Aviation regulatory bodies have already brought out that network security should be identified and needs to be addressed. Security issues in an Avionics Wireless Network (AWN) and a modified Advanced Encryption Standard (AES) algorithm, based on time synchronisation, is brought out in the paper.

Keywords—Wireless, security, AWN, Avionics, AES, Key Management, Synchronization

I. INTRODUCTION

Improving the performance of an aircraft has been the primary aim of the aircraft designers both for commercial and military aircraft. This translates to increased weapon or load carrying capacity and providing the crew with faster and updated information. The capability to carry additional load would require a reduction in the overall weight of the aircraft. Aircraft designers have identified wireless as a suitable option to reduce the weight of the aircraft. In addition, wireless can resolve issues related to aging of onboard wires, EMI/EMC effect due to wires, increased downtime due to faulty wires, installation and maintenance cost of wires. Security in wireless systems was one of the key reasons that prevented designers to implement wireless technology for flight critical systems. Security issues in the Wi-Fi in-flight entertainment networks of A350 and A380 and their vulnerability to hacking was highlighted by Federal Aviation Administration (FAA) in the reports submitted in 2008 and again in 2015.

The security issues in AWN are likely to be like those experienced by Wireless Sensor Networks (WSN). The aim of an attacker is either to interrupt, interceptor modify the data between two nodes. The attacks on sensor networks are brought out in [1,2,3] and those likely to be faced by onboard wireless networks are discussed in [4,5]. Some of these attacks are discussed below.

1) *Replay attack*: The attacker resends a message, from the network, to the original device to create infinite loops and increase end to end latency. Increased latency will severely affect critical systems like auto pilot, engine etc. *Routing loops attack* is also like replay attack, in which data packets are routed through the same routers continuously.

2) *Spoofing attack*: Spoofing attack may involve ARP spoofing or MAC spoofing. Spoofing attacks can lead to data theft and deletion of data from the network. It can also result in DoS.

3) *Selective Forwarding*: Also known as the drop data packet attack, the compromised node deliberately drops certain data packets. This is used as an effective method in multi-hop scenarios. In an AWN, this type of attack can be catastrophic, if a compromised node drops few data packets related to control system.

4) *Man in the Middle*: In this type of attack, a secure connection is established by the attacker with both the sending and receiving nodes. Both the nodes treat the new node in the middle as a genuine node. Attacker can use this type of attack to modify the navigation data and control the aircraft.

5) *Node Compromise*: In an aircraft, multiple teams are working for maintenance. Therefore there is a high possibility of an attacker taking physical control over a node. Once a node is compromised, *sinkhole* type of attack to route the traffic towards the compromised node can take place.

6) *Denial of Service (DoS) Attack*: The attack occurs at the physical level and is one of the most severe attacks and make the network inaccessible to system LRUs. Radio jamming is also a type of DoS attack, which interfere with the network protocol and also cause battery exhaustion.

7) *Eavesdropping*: An adversary can do eavesdropping and listen to the communication on the channel and access valuable information to disrupt the network using specific attacks.

The paper discusses key management and time synchronization aspects for implementing a wireless network for aircraft intra communication. A modified AES encryption scheme based on time synchronized LRUs (Line Replaceable Units) for AWN is proposed in the paper.

II. KEY MANAGEMENT AND TIME SYNCHRONIZATION IN WIRELESS NETWORK

Two important aspects for security in distributed wireless networks are key management and time synchronization. The proposed technique for AES modification, discussed later in the paper, is based on key modification and time synchronization of sending and receiving units with the aircraft system clock.

A. Key management

Key management in WSN or AWN will involve the following tasks:

- Key generation and distribution to the network nodes.
- Key establishment between communicating nodes, which may either be a centralized key or a session key. In an AWN, the key establishment will be simpler since deployment of LRUs is known.
- Key revocation and issue of fresh keys, if a node is compromised.
- Key update is regularly required in a session-based key management.

Three different schemes for key management in wireless networks, which can be adopted for AWN, are discussed below.

1) *Pre-distributed Master-Key Scheme*[6]: In this scheme, it is assumed that all nodes in a set S are trustworthy. The authentication is implemented by preloading two master keys, group authentication key and key generation key. A pair of sensor nodes from the set S authenticates each other using the authentication key. Once authentication keys are exchanged, the nodes then establish session keys using a master key. However, access to the master key can make the whole sensor network insecure.

2) *Pre-distributed Random Key Scheme* [7]: Each sensor node initially selects a pool of m keys (node's key ring) out of possible S keys. Any two random subsets, having m keys each, will have at least one key in common. During the key establishment phase, communicating nodes identify one common key (from m keys), which is used as their shared key. Modifications have been proposed to increase the common keys, to increase the strength of the network against attacks. However, if the keys of the total pool S are compromised, the complete network is compromised.

3) *Deployment based Key Management Scheme*[8]: In some sensor networks, the deployment information of the nodes is known. The scheme has three phases namely key pre-distribution, shared-key discovery, and path-key establishment. In the first phase, key pool S is divided into a set of key pools. Based on the deployment knowledge, neighbouring groups have neighbouring key pools that have greater number of common keys. As the groups move farther away from each other, the common keys keep reducing. From these neighbouring key pools, m keys are selected and the process continues as in [7].

B. Time Synchronization

All devices on a network should be time-synchronized to a central clock to undertake distributed tasks like sensor data fusion, cryptographic schemes, coordination of scheduled tasks, etc. However, two nodes of a network can be out of synchronisation due to Clock Offset, Clock Skew or Drift Error. Additional factors, that affects time synchronization in a wireless network includes Send Time, Receive time, Propagation Time and Access Time. Network Time Protocol (NTP) has been providing time synchronization for network devices based on the time stamp information. With more stringent requirements, new techniques have been proposed for time synchronization. Three approaches for time synchronization are discussed below.

1) *Reference Broadcast Synchronization (RBS)* [9]: RBS uses the receiver to receiver synchronization. A third node will broadcast a beacon to both the receivers. The receivers use the arrival time of the beacon as a reference point. This has the advantage of the sender not being part of the synchronization system and thus provides an advantage over NTP.

2) *Timing-sync Protocol for Sensor Networks (TPSN)* [10]: TPSN protocol has two steps for a unique time scale across the network. In the first step, each node is assigned a level. The central computer/server is assigned node level 0. The node at level n can communicate with at least one node at level $n-1$. In the second phase, pair wise synchronization is performed. During this phase, a node at level n synchronizes to a node at level $n-1$. Finally, all nodes in the network synchronize their clocks to the node at level 0.

3) *The Flooding Time Synchronization Protocol (FTSP)* [11]: FTSP uses the concept of MAC layer time stamping. The first step is the selection of a root node. The time synchronization messages are originated by the root node. A node's clock is adjusted whenever it receives a time synchronization message from the root node. The node then broadcasts its own time to its neighbors for synchronization. Thus all the nodes are synchronized to the root node.

C. Key Management and Time Synchronisation in AWN

In wired communication onboard the aircraft, key management was not an issue. However, timing synchronization has always been a critical issue due to synchronization requirements between different onboard devices and sensors. During the powering up of the LRUs (Line Replaceable Unit) of an aircraft, the first activity is the synchronization of the LRUs based on the aircraft time server. The onboard time server maintains its clock using GPS time or the Inertial Navigation System (INS) time. Clock synchronization in aircraft networks is usually carried out using the NTP with an accuracy of 1-10 msec. Inter-Range Instrumentation Group (IRIG) provides time accuracy of 1-10 μ S and is mostly used in mission-critical operations like space and military. With the introduction of Precision Time Protocol (PTP), defined in the IEEE-1588 standard, the accuracy can be improved to 20-100nS and is cost-effective due to the use of Ethernet as the physical medium.

In AWN, the location of units is fixed, and so key management is much simpler compared to WSN. Therefore, in an AWN, deployment based key management scheme which uses the concept of pre-distributed random key schemes [7,8] can be used. FTSP is the most suitable time synchronisation protocol for AWN. Mission computer (MC) can be the root node and all nodes can be synchronized to the MC. In the proposed modified AES, discussed later in the paper, time synchronization between sender and the receiver plays an important role.

D. Security Algorithms

The security algorithms for WSN include symmetric and asymmetric cryptography. Under the symmetric key algorithm, only a single key is used for encryption and decryption. The most common symmetric algorithms available for encryption are the AES, DES, 3DES, Blowfish, Twofish, etc. Asymmetric cryptography uses different keys for encryption and decryption. Also known as public-key cryptography, the two keys include a public key and a private key. Algorithms using asymmetric cryptography include RSA, DSA, Elliptic curve techniques, etc. Some of the most popular encryption schemes are

1) *DES/3DES*: Data Encryption Standard (DES) was published in 1970 and uses a 64-bit block. Only 56 bits are used for encryption and eight bits are used for parity. The key length is fixed, and the number of rounds is 16. 3DES uses the same algorithm as the DES. However, 3DES uses encryption, decryption and again encryption using 3 different keys. So effectively 3DES uses a 168-bit key.

2) *Twofish*: Twofish algorithm was proposed by Bruce Schneier, John Kelsey, Doug Whiting, David Wagner. The algorithm accepts a variable key length of 128bits, 192 bits and 256 bits. It is also a 16 rounds algorithm and uses key-dependent four 8 X 8 bits S-boxes and a fixed 4 X 4 maximum distance separable matrix.

3) *RSA*: Rivest, Shamir, Adleman (RSA) algorithm was developed in 1977. It generates two keys, the public key, and the private key. The main operation of RSA involves the selection of two random prime numbers and modular exponentiation.

4) *AES[12]*: The most popular encryption scheme being used is Advanced Encryption Standard (AES). AES was the result of the National Institute of Standards and Technology's (NIST) call for block cipher after DES was cracked. The encryption algorithm *Rijndael* developed by J. Daemen & V. Rijmen was selected as the AES. The number of rounds 10, 12, 14 in AES is dependent on the key size of 128, 192 or 256 bits respectively. A block diagram of AES algorithm is shown in Fig. 1. Four functions, namely SubByte transformation, ShiftRow, MixColumn, and AddRoundKey constitute one round of encryption. During decryption the same process is carried out in the reverse order i.e. AddRoundKey, MixColumn, ShiftRow, SubByte transformation.

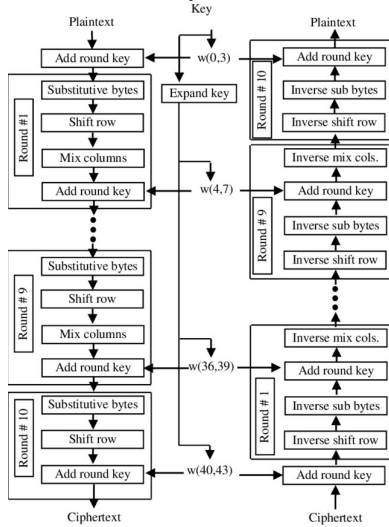


Fig. 1. AES Block Diagram

III. MODIFIED AES FOR AWN

Researchers have proposed various modifications to AES [13], [14], [15]. This paper presents modifications to AES by generating the key and S-box based on the aircraft system clock. The LRUs can be synchronized with the MC using the FTSP time synchronization method. Multiple levels of randomness is introduced, including time based key, S-box, modification of the new S-box and generating a new inverse S-box for each session to prevent replay attacks and man in the middle attacks. This will eliminate the need to share the new key between the sender and receiver for each session and thus prevent unauthorized access to the key.

A. Procedure for varying key and S-box for each session

The steps involved in the proposed technique, for modified AES, at the transmitting and receiving LRUs are indicated in Fig. 2.

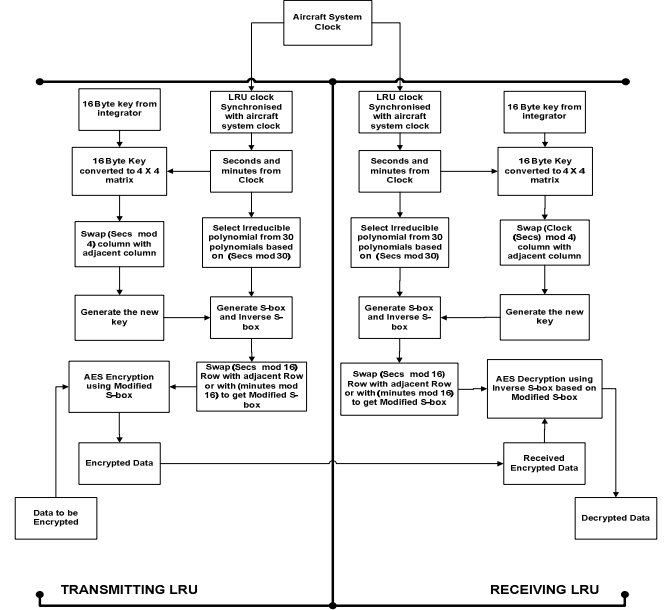


Fig. 2. Block diagram of modified AES

- *Step 1*: The 16 Byte key is transformed into a 4X4 matrix.
- *Step 2: ColumnSwap*: Based on the seconds from the clock, $\{(seconds) \bmod 4\}^{th}$ column and adjacent column of the key matrix are swapped.
- *Step 3*: Transform the 4X4 matrix back to a 16 Byte key, to obtain a new key.
- *Step 4*: Generation of S-box: The AES algorithm uses the concept of polynomial multiplication modulo using an irreducible polynomial of degree 8 to generate the S-box. AES standard specifies the irreducible polynomial as $m(x)=x^8+x^4+x^3+x+1$. The proposed technique uses 30 different possible irreducible polynomials of degree 8 [16]. Based on $\{(seconds) \bmod 30\}$, one out of the 30 polynomials is selected. The S-box for encryption is generated based on the selected irreducible polynomial.
- *Step 5: RowSwap*: The generated S-box (16X16) is modified again based on $\{(seconds) \bmod 16\}$. Two adjacent rows of the S-box based on $\{(seconds) \bmod 16\}$ are swapped. This generates a new S-box.
- *Step 6*: During decryption inverse of the modified S-box is used. Since both the LRUs are time synced, the S-box and inverse S-box at both the units are the same.

IV. RESULTS AND DISCUSSIONS

The simulations, for two different sessions using the proposed technique, have been carried out in MATLAB. The shared key available at the LRU, is assumed to be the same in all cases of simulation and is taken as [00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f]. In actual implementation, the shared key will vary depending on the key management scheme. For each change in time (sec), a modified key is generated. The new key for each session is used for encryption. During the same session, a new irreducible

polynomial (out of 30 polynomials) is selected, based on the time (sec). Using this irreducible polynomial, a new S-box, for *SubBytes Transformation* is generated, for encryption and correspondingly, during decryption, an inverse S-box for *InvSubBytesTransformation* is generated. The generated S-box and the inverse S-box are modified based on the time (sec).

A. Session 1

Key= [00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f]

Step1:

00	04	08	0c
01	05	09	0d
02	06	0a	0e
03	07	0b	0f

	Year	Mon	Day	Hr	Min	Sec
Clock	2020	2	1	5	34	3.29200

Clock Seconds=4

TABLE I. GENERATED S-BOX FOR SESSION 1

99	73	3	35	38	250	67	156	180	54	218	167	6	15	233	235
124	49	179	158	244	190	214	68	234	123	51	59	206	44	239	164
153	28	74	140	126	42	232	150	168	113	141	39	204	17	240	13
48	159	229	11	50	145	138	105	248	129	86	162	72	237	200	40
30	208	220	7	130	100	225	33	152	45	199	216	211	185	236	253
173	64	176	55	115	29	184	170	89	154	157	108	10	166	245	148
202	238	104	16	32	96	34	187	203	207	26	117	151	144	102	2
181	95	155	197	19	217	77	57	196	241	61	22	137	70	247	41
221	9	186	118	201	0	81	114	147	161	149	43	87	5	66	101
119	188	27	136	71	1	109	191	80	85	121	209	21	82	134	83
4	65	135	231	255	20	60	94	107	212	92	189	251	172	242	98
120	111	143	210	58	79	110	75	142	177	88	192	14	128	228	37
183	163	165	171	230	25	175	133	194	93	226	76	195	112	122	56
205	106	103	243	12	52	252	97	132	47	90	193	227	84	222	223
8	24	125	174	31	215	69	63	46	254	62	169	116	23	78	91
127	18	224	219	178	246	139	249	146	36	213	131	53	198	160	182

TABLE II. MODIFIED S-BOX AFTER ROWSWAP

99	73	3	35	38	250	67	156	180	54	218	167	6	15	233	235
124	49	179	158	244	190	214	68	234	123	51	59	206	44	239	164
153	28	74	140	126	42	232	150	168	113	141	39	204	17	240	13
30	208	220	7	130	100	225	33	152	45	199	216	211	185	236	253
48	159	229	11	50	145	138	105	248	129	86	162	72	237	200	40
173	64	176	55	115	29	184	170	89	154	157	108	10	166	245	148
202	238	104	16	32	96	34	187	203	207	26	117	151	144	102	2
181	95	155	197	19	217	77	57	196	241	61	22	137	70	247	41
221	9	186	118	201	0	81	114	147	161	149	43	87	5	66	101
119	188	27	136	71	1	109	191	80	85	121	209	21	82	134	83
4	65	135	231	255	20	60	94	107	212	92	189	251	172	242	98
120	111	143	210	58	79	110	75	142	177	88	192	14	128	228	37
183	163	165	171	230	25	175	133	194	93	226	76	195	112	122	56
205	106	103	243	12	52	252	97	132	47	90	193	227	84	222	223
8	24	125	174	31	215	69	63	46	254	62	169	116	23	78	91
127	18	224	219	178	246	139	249	146	36	213	131	53	198	160	182

B. Session 2

Key= [00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f]

Step1:

00	04	08	0c
01	05	09	0d
02	06	0a	0e
03	07	0b	0f

	Year	Mon	Day	Hr	Min	Sec
Clock	2020	2	1	5	35	37.43000

Clock Seconds=38

Step2: ColumnSwap=(38 mod 4)=2, (2nd and 3rd columns of key matrix are swapped)

Step2: ColumnSwap=(4 mod 4)=0+1, (1st and 2nd (adjacent column) of key matrix are swapped)

04	00	08	0c
05	01	09	0d
06	02	0a	0e
07	03	0b	0f

Step 3: New Key=[04 05 06 07 00 01 02 03 08 09 0a 0b 0c 0d 0e 0f]

Step 4: Irreducible Polynomial No= (4mod30) +1=5

Selected Irreducible Polynomial= "100111001"

The generated S-box is shown in Table I. The generated S-box is modified using RowSwap

Step 5: RowSwap= (4 mod 16) =4(4th and 5th rows of S-box are swapped)

The modified S-box after RowSwap is shown Table II. A new inverse S-box is generated from the modified S-box.

00	08	04	0c
01	09	05	0d
02	0a	06	0e
03	0b	07	0f

Step 3: New Key=[00 01 02 03 08 09 0a 0b 04 05 06 07 0c 0d 0e 0f]

Step 4: Irreducible Polynomial No= (38mod30) +1=9

Selected Irreducible Polynomial= "101100011"

The generated S-box (Table III) is modified using RowSwap.

Step 5: RowSwap= (38 mod 16) =6(6th and 7th rows of S-box are swapped).

The modified S-box after RowSwap is given in Table IV. Inverse S-box is generated from the modified S-box

TABLE III. S-BOX FOR SESSION 2

99	251	34	29	195	182	92	85	51	113	137	127	252	76	120	38
124	206	27	23	219	129	245	158	78	8	147	93	186	208	130	57
225	233	184	116	82	17	89	153	63	170	18	111	40	209	144	201
96	143	22	14	106	79	75	86	204	193	238	97	244	146	172	194
47	46	43	241	131	248	229	95	246	223	87	215	115	250	30	224
98	13	165	49	50	177	185	5	179	112	132	103	155	36	168	142
226	197	24	236	217	110	213	10	231	202	117	148	122	11	249	178
104	83	214	81	138	2	64	55	54	26	154	247	58	12	198	94
72	133	200	15	71	65	39	243	19	59	163	160	32	128	125	60
108	77	157	140	234	123	6	44	152	166	139	56	205	4	31	181
227	70	84	1	0	16	74	126	203	187	7	25	3	189	80	145
52	102	121	171	48	150	169	192	175	220	221	191	162	237	107	69
174	161	61	88	211	228	164	199	62	136	232	105	53	100	218	28
41	167	159	67	210	141	196	156	253	73	188	37	212	173	207	242
230	21	118	42	180	109	119	135	151	9	222	114	255	66	68	216
149	235	101	176	183	91	33	20	134	190	35	240	90	254	239	45

TABLE IV. MODIFIED S-BOX AFTER ROWSWAP

99	251	34	29	195	182	92	85	51	113	137	127	252	76	120	38
124	206	27	23	219	129	245	158	78	8	147	93	186	208	130	57
225	233	184	116	82	17	89	153	63	170	18	111	40	209	144	201
96	143	22	14	106	79	75	86	204	193	238	97	244	146	172	194
47	46	43	241	131	248	229	95	246	223	87	215	115	250	30	224
226	197	24	236	217	110	213	10	231	202	117	148	122	11	249	178
98	13	165	49	50	177	185	5	179	112	132	103	155	36	168	142
104	83	214	81	138	2	64	55	54	26	154	247	58	12	198	94
72	133	200	15	71	65	39	243	19	59	163	160	32	128	125	60
108	77	157	140	234	123	6	44	152	166	139	56	205	4	31	181
227	70	84	1	0	16	74	126	203	187	7	25	3	189	80	145
52	102	121	171	48	150	169	192	175	220	221	191	162	237	107	69
174	161	61	88	211	228	164	199	62	136	232	105	53	100	218	28
41	167	159	67	210	141	196	156	253	73	188	37	212	173	207	242
230	21	118	42	180	109	119	135	151	9	222	114	255	66	68	216
149	235	101	176	183	91	33	20	134	190	35	240	90	254	239	45

C. Avalanche test

Avalanche test was carried out to compare the performance of AES and the proposed technique. The same key *[00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f]* was used for AES and the proposed technique. The two inputs of 128 bit, with a variation of a single bit, provided to both encryption techniques are

Input 1: *[0011 22 33 44 55 66 77 88 99 aa bb cc dd ee ff]*

Input 2: *[01 11 22 33 44 55 66 77 88 99 aa bb cc dd ee ff]*

TABLE V. AVALANCHE TEST RESULTS

Input	AES		Proposed Technique	
	Output bit changes	Avalanche effect	Output bit changes	Avalanche effect
128	64	0.5	69	0.5390
128	61	0.4765	62	0.4843
128	62	0.4843	69	0.5390
128	63	0.4921	61	0.4765
128	60	0.4687	66	0.5156
Average Avalanche effect		0.4844		0.5109

The avalanche test results obtained for AES and proposed technique encryption is given in Table V. The results indicate that the proposed technique results in change in more than 50% bits at the output with a single bit change at the input and the results of the proposed technique are better than that of AES.

D. Execution Time

A comparison of execution time was carried out, on an Intel i5, 1.6 GHz machine between AES and the proposed technique. The execution time for both the techniques is given in Table VI. Fig. 3 gives the comparison of the execution time for the two encryption techniques. The results indicate that execution time for the proposed technique is

slightly more than the AES execution time, due to the additional computations pertaining to modifications suggested in the proposed technique.

TABLE VI. EXECUTION TIME RESULTS

Data Size (Bits)	AES	Proposed Technique
1000	1.62159	2.17673
2000	2.513072	3.028276
3000	3.528054	4.029843
4000	4.73941	5.030824
5000	5.831872	5.983887
6000	6.475931	7.092577
7000	8.071796	9.033346
8000	9.577389	10.68863
9000	10.88382	11.67851
10000	12.13289	13.01404

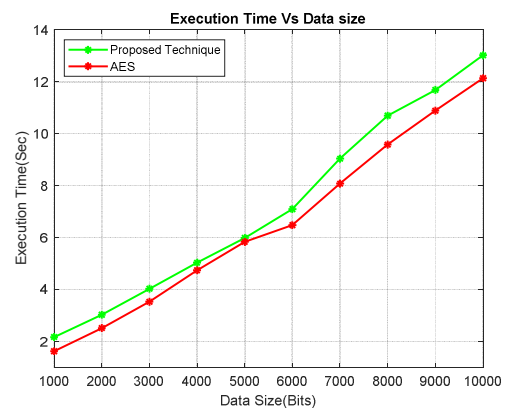


Fig. 3. Execution time comparison

E. Replay attack

In order to test the efficacy of the proposed technique, a replay attack was carried out and the results is shown in

```

Clock_sec =
38

in =
'S<fla1c[j]'xiNIJ=PPmk)CmRAYsT[W3<L5p2505I=wHlu)I*8FX8Yc5+<>HP(81"xenV6K(SQ6N['EB]
#tvk)8?',d"Njcut7b2'

out =
'S<fla1c[j]'xiNIJ=PPmk)CmRAYsT[W3<L5p2505I=wHlu)I*8FX8Yc5+<>HP(81"xenV6K(SQ6N['EB]
#tvk)8?',d"Njcut7b2'

Clock_sec_attack =
49

out_attack =
'S<8CUCU&4ih 000 ?I Fa5aa8 . >B "ho%*-4r0d2 8h8Jm t p|p6 E%At040000000R c <W
&MI-v-Ad
qH6 :ip8 *v. 4itJ1880i8J1884!'

```

F. Analysis

V. CONCLUSION

synchronized time at the two LRUs. The time is used for generating session based keys, selecting an irreducible polynomial, generating S-box and inverse S-box, that keeps changing for each session and thus improves the robustness of the encryption algorithm.

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